1.) A collection of interconnected computing devices that communicate with each other to share resources, data, and services.

2.)

Routers- enable computers to communicate and they can pass information between two networks such as between home network and the Internet.

Hubs- enable computers on a network to communicate. Each computer plugs into the hub with an Ethernet cable, and information sent from one computer to another passes through the hub.

Access Points- provide wireless access to a wired Ethernet network. An access point plugs into a hub, switch, or wired router and sends out wireless signals. This enables computers and devices to connect to a wired network wirelessly.

Switches- work the same way as hubs, but they can identify the intended destination of the information that they receive.

3.) Transmission media is the physical pathway or communication channel that carries data from a transmitter to a receiver.

Guided Media: Uses physical cables.

- Media Types: Twisted Pair cables, Coaxial cables, Fiber Optic cables.
- Signal Types*:* Electrical signals and Light pulses.

Unguided Media: Uses the air to transmit data without physical boundaries.

- Media Types: Radio waves, Microwaves, Infrared.
- Signal Types: Electromagnetic waves.

4.) UTP (Unshielded Twisted Pair): Contains twisted pairs of copper wires without any additional metallic shielding. Cheaper, More flexible, Easy to installation, but more susceptible to Electromagnetic Interference (EMI) and crosstalk.

STP (Shielded Twisted Pair): Contains twisted pairs wrapped in metallic shield. This shielding significantly reduces EMI and crosstalk, making it suitable for environments with heavy electrical interference, but it is thicker and more expensive.

5.) Topology refers to the physical or logical arrangement of the nodes and connections in a network.

I. Star Topology, Bus Topology, Ring Topology, Mesh Topology

II.

Topology	Advantages	Disadvantages
Star	failure of one node doesn't affect the rest	If the central hub fails, the entire network goes down
Bus	Very cheap and easy to set up for small networks	If the main cable fails, the network goes down
Ring	no central hub needed	Failure of one node or cable break can take down the whole network
Mesh	Highly reliable and robust	Very expensive and complex to cable and configure

6.) Intranet- A private, secure network used to share company information and computing resources among employees.

Extranet- A controlled private network that allows authorized external users access to part of an organization's intranet.

7.) Command- copy running-config startup-config

Mode- Privileged EXEC Mode (Router#)

8.)

Router#configure terminal

This command moves the user from Privileged EXEC mode into Global Configuration mode, allowing them to make changes that affect the whole router.

Router(config)# enable password abc

This sets the unencrypted password required to enter Privileged EXEC mode. The user will now need to type "abc" when using the enable command.

9.) Router# show running-config

10.)

```
Router>enable
Router#show ?
  aaa                 Show AAA values
  access-lists        List access lists
  arp                 Arp table
  cdp                 CDP information
  class-map           Show QoS Class Map
  clock               Display the system clock
  controllers         Interface controllers status
  crypto              Encryption module
  debugging           State of each debugging option
  dhcp                Dynamic Host Configuration Protocol status
  dot11               IEEE 802.11 show information
  ephone              Show all or one ephone status
  file                Show filesystem information
  flash:              display information about flash: file system
  frame-relay         Frame-Relay information
  history             Display the session command history
  hosts               IP domain-name, lookup style, nameservers, and host table
  interfaces          Interface status and configuration
  ip                  IP information
  ipv6                IPv6 information
  line                TTY line information
  logging             Show the contents of logging buffers

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show version
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1,
 RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

ROM: System Bootstrap, Version 12.3(8r)T8, RELEASE SOFTWARE (fc1)

System returned to ROM by power-on
System image file is "flash:c1841-advipservicesk9-mz.124-15.T1.bin"


This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
```

```
Router#show protocol
Global values:
  Internet Protocol routing is enabled

Router#show flash

System flash directory:
File  Length   Name/status
  3   33591768 c1841-advipservicesk9-mz.124-15.T1.bin
  2   28282    sigdef-category.xml
  1   227537   sigdef-default.xml
[33847587 bytes used, 30168797 available, 64016384 total]
63488K bytes of processor board System flash (Read/Write)


Router#show running-config
Building configuration...

Current configuration : 489 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
!
!
!
!

Router#show history
  router>
  >
  disable
  enable
  configure terminal
  show version
  show protocol
  show flash
  show running-config
  show history
Router#show clock
*0:22:48.53 UTC Mon Mar 1 1993
```

```
Router#show hosts
Default Domain is not set
Name/address lookup uses domain service
Name servers are 255.255.255.255

Codes: UN - unknown, EX - expired, OK - OK, ?? - revalidate
       temp - temporary, perm - permanent
       NA - Not Applicable None - Not defined

Host                      Port  Flags      Age Type   Address(es)
Router#show users
    Line       User       Host(s)              Idle       Location
*  0 con 0                idle                 00:00:00

  Interface    User               Mode         Idle     Peer Address
Router#show interfaces
FastEthernet0/0 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 00d0.d374.a901 (bia 00d0.d374.a901)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 1 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out

Router#show protocols
Global values:
  Internet Protocol routing is enabled
 --More--
```